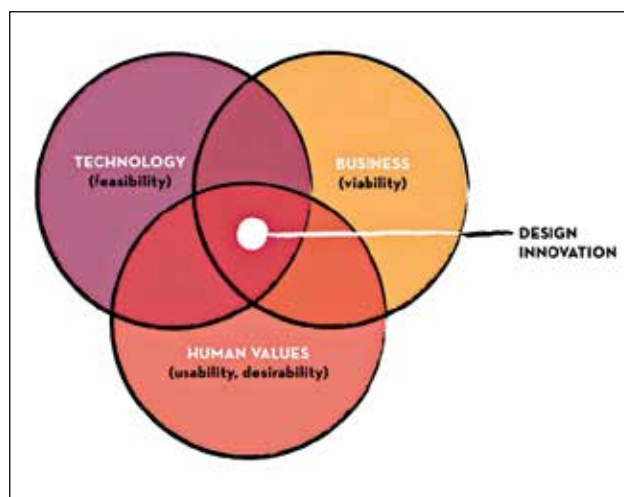


there was an attempt to solve problems but during the industrial revolution, more efficient solutions were the need of the hour. This was a time when great minds applied themselves to finding solutions to problems that people didn't even realise they had. This attitude in design continues till date as mobile application designers try to make applications more usable for the end-user, applications that have existed much before their time. In the 1960s, these ideas began to be documented and taught. The idea then was that Design Thinking would help its learners share ideas on how to successfully approach and solve difficult, multi-dimensional problems, and more specifically understand effective methods to ideate, select and execute solutions that individuals and businesses could use to improve their own problem solving processes and skills.

Design thinking has come to be defined as combining empathy (for the context of a problem), creativity (in the generation of insights and solutions) and rationality (in analysing and fitting various solutions to the problem context). Design thinking is a formal method for practical, creative resolution of problems and creation of solutions, with the intent of an improved future result. In this regard, it's a form of solution-focused thinking; starting with the goal of a better future situation instead of solving a specific problem. This, in turn, makes it difficult to describe as just another scientific method of analysing a given problem, which begins with thoroughly defining all the parameters of the problem in order to create a solution and ends with a working solution. Design Thinking identifies and investigates with both, known and ambiguous aspects of the current situation in order to discover hidden parameters and open alternative paths which may lead to the goal. Design Thinking is iterative, and intermediate "solutions" are treated as potential starting points of alternative paths, even if it means redefining the initial problem. It's a means of innovative thinking when scientific thinking fails to deliver.

The goal of design thinking is to solve a problem using both, analysis and synthesis iteratively, which essentially means breaking a problem into various solutions (analysis) and then rejecting solutions based on given



'Design Thinking' can be described as a discipline that uses the designer's sensibility and methods to match people's needs with what is technologically feasible and what a viable business strategy can convert into customer value and market opportunity – Tim Brown, CEO of IDEO

parameters and constraints to find an efficient solution (synthesis) at each step as opposed to finding a straight forward solution; making the solution adaptable to future changes in the problem. Analysis and Synthesis, as scientific methods, always go hand in hand; they complement one another. Every synthesis is built upon the results of a preceding analysis, and every analysis requires a subsequent synthesis in order to verify and correct its results. Considering alternative and intermediate solutions makes this a process of design thinking. Design Thinkers also use 'divergent thinking' and 'convergent thinking' to explore many possible solutions. Divergent Thinking is the kind of thinking that offers different, unique or

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variant ideas adherent to one theme and is seen as a creative process, while Convergent Thinking is the kind of thinking that focuses on the most promising solution to a given problem and is thus more of an analytical process. Design thinking encourages divergent thinking to ideate many solutions (possible or impossible) and then uses convergent thinking to prefer and realise the best resolution. While a scientific solution is effective for problems with unknown outcomes, design thinking advocates considering known to formulate methods to predict outcomes, ultimately finding an efficient solution.

Designers communicate in visual or object language. Symbols, signs, and metaphors are used through the medium of sketches, diagrams and technical drawings to translate

abstract requirements into concrete objects. The way designers communicate, then, is through understanding this way of coding design requirements in order to produce built products. John Chris Jones, a Design Science theorist, responded to 'Is designing an art, a science or a form of mathematics?' – "The main point of difference is that of timing. Both, artists and scientists operate in the physical world as it exists in the present (whether it's real or symbolic), while mathematicians operate on abstract relationships that are independent of historical time. Designers, on the other hand, are forever bound to treat as real that which exists only in an imagined future and have to specify ways in which the foreseen thing can be made to exist."

Design can be seen as its own culture in education, with its own methods and ways of thinking which can be systematically taught across both, K-12 and higher education. Intel® has a design thinking for educators programme – the details of which can be found at <http://dgit.in/1s3o3u1>. Intel® has also been trying to bring K-12 education to India; read more about it at <http://dgit.in/1qsPLyr>.

HOW DO I THINK DESIGN?

1. FOLLOW BASIC RULES

- i. **Human Rule:** Measure your solution in a group of solutions.
- ii. **Re-design Rule:** Solve the solved problem to find a better solution.